FLOW CONTROL AT DATA LINK LAYER USING SLIDING WINDOW PROTOCOL

AIM:

To write the program to implement flow control at the Data link layer using the sliding window protocol, and to simulate the flow of frames from one node to another.

ALGORITHM:

- Start the program
- Read that frames N, window size w, & lost frame
- Intialize base -0 and next seq -0
- Repeat until all frames are transmitted
  - create the current sender windows
  - Send all frames in the window
  - If a frame is lost, (Go-Bak-N)
  - otherwise receive ACKs & slide the window

forward

- Repeat until all frames are successful
- Stop.

CODE:

```
def simulate (n, w, lost):
    base = 0
    while base < n:
        end = min (base + w, n)
        print (f "In window : [{base} - {end-1}]")
        for i in range (base, end)
            print (f" A -> Sending Frame {i}")
```

```
=== SLIDING WINDOW PROTOCOL - FLOW CONTROL SIMULATION (Go-Back-N) ===

Enter total number of frames to send       : 10
Enter sender's window size                 : 3
Enter frame number to simulate as LOST (0 to 9, enter -1 for no loss): 2

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Sender Window : [Frame 0 - Frame 2]
Node A --> Sending Frame 0
Node A --> Sending Frame 1
Node A --> Sending Frame 2
        (Frame 2 LOST in transit!)
Node B --> Frame 2 missing. Discarding out-of-order frames after it.
Node A --> Timeout! Resending window starting from Frame 2.

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Sender Window : [Frame 0 - Frame 2]
Node A --> Sending Frame 2
Node B --> Received Frame 0, sending ACK 0
Node B --> Received Frame 1, sending ACK 1
Node B --> Received Frame 2, sending ACK 2
Node A --> ACK 2 received. Sliding window forward.

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Sender Window : [Frame 3 - Frame 5]
Node A --> Sending Frame 3
Node A --> Sending Frame 4
Node A --> Sending Frame 5
Node B --> Received Frame 3, sending ACK 3
Node B --> Received Frame 4, sending ACK 4
Node B --> Received Frame 5, sending ACK 5

Node A --> ACK 5 received. Sliding window forward.

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Sender Window : [Frame 6 - Frame 8]
Node A --> Sending Frame 6
Node A --> Sending Frame 7
Node A --> Sending Frame 8
Node B --> Received Frame 6, sending ACK 6
Node B --> Received Frame 7, sending ACK 7
Node B --> Received Frame 8, sending ACK 8
Node A --> ACK 8 received. Sliding window forward.

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Sender Window : [Frame 9 - Frame 9]
Node A --> Sending Frame 9
Node B --> Received Frame 9, sending ACK 9
Node A --> ACK 9 received. Sliding window forward.

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All 10 frames successfully transmitted from Node A to Node B.
```

```
        if f == lost:
            print(f"Frame {f} LOST")
            print(f"B → Frame {i} missing")
            print(f"A → Timeout, resending from {f}...")
            lost = -1
            break
    else:
        for i in range(base, end):
            print(f"B → Received Frame {f}, ACK {f}")
        base = end
print("\n All frames transmitted successfully")
n = int(input("Total frames:"))
w = int(input("window size:"))
lost = int(input("Lost frame (-1 for none):"))
simulate(n, w, lost)
```

RESULT:

Thus, the sliding window Protocol using Go-Back-N was successfully implemented.

